

DEPARTMENT OF CSE
PROJECT ABSTRACT SUBMISSION
COURSE: DATA ANALYTICS AND VISUALIZATION – 24DSB3101
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TITLE OF THE PROJECT

Intelligent Customer Churn Prediction Using Data Analytics and Classification Models

Student Details

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1.Title:

Intelligent customer churn prediction using data analytics and classification models

2.Dataset

This project uses publicly available, widely referenced customer churn datasets so that results can be reproduced and benchmarked against existing literature. Two complementary datasets are used to cover both the telecommunications and banking domains.

2.1 IBM Telco Customer Churn Dataset

A widely used benchmark dataset representing a fictional telecommunications company providing phone and internet services to 7,043 customers in California. It contains 21 attributes covering customer demographics, account information, subscribed services, and churn status, and is one of the most frequently cited datasets in churn-prediction research.

Key Attributes: customerID, gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetService, OnlineSecurity, TechSupport, StreamingTV, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, Churn.

Source: Kaggle (originally distributed by IBM Cognos Analytics sample data)

Dataset Link: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

2.2 Bank Customer Churn Dataset (Churn Modelling)

A widely used banking-sector churn dataset containing 10,000 customer records from a European bank, with 13 attributes describing demographic, account, and engagement information along with a binary churn label (Exited).

Key Attributes: CustomerId, Surname, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, Exited.

Source: Kaggle (contributed by M. Akturk)

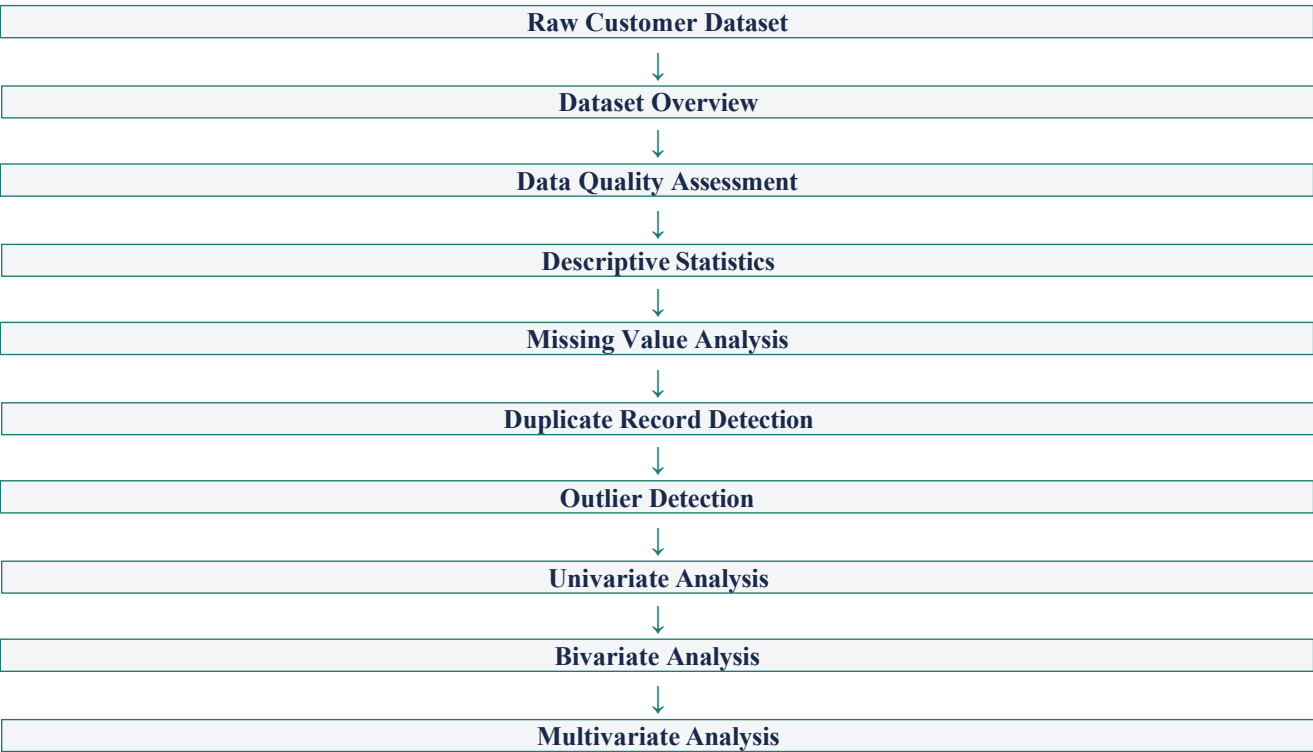
Dataset Link: <https://www.kaggle.com/datasets/mathchi/churn-for-bank-customers>

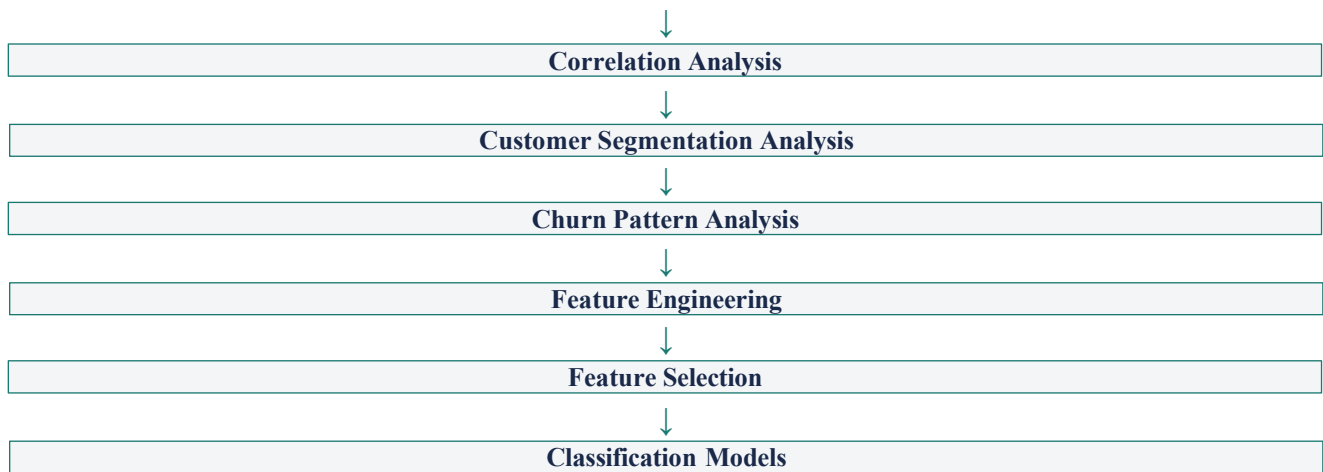
3. Abstract

Customer retention has become a central determinant of long-term profitability across telecommunications, banking, insurance, subscription, and e-commerce businesses, since acquiring a new customer is well established to be considerably more expensive than retaining an existing one. This project addresses the problem of customer churn by developing a structured, data-driven analytics pipeline that examines customer demographic, behavioral, service-usage, and billing data to identify individuals who are likely to discontinue their relationship with an organization. The methodology begins with systematic data preprocessing, including missing-value treatment, duplicate handling, categorical encoding, feature scaling, and class-imbalance correction, followed by a comprehensive **Exploratory Data Analysis (EDA)** covering univariate, bivariate, multivariate, and correlation analysis to understand the relationships between customer attributes and churn behavior. Guided by these insights, targeted **Feature Engineering** is applied to construct behaviorally meaningful indicators such as tenure groups, charge ratios, and service-usage combinations. Multiple **Classification Models** — Logistic Regression, Decision Tree, Random Forest, XGBoost, and CatBoost — are trained on the processed data and evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC, with particular attention to metrics beyond Accuracy given the class imbalance typically present in churn datasets. The comparative results are consolidated into an interactive analytics dashboard that visualizes churn distribution, feature importance, and high-risk customer segments, translating model output into insights a non-technical stakeholder can act upon. Through this combination of **Predictive Analytics** and interpretable visualization, the project aims to help organizations identify at-risk customers early and design targeted, evidence-based retention strategies that reduce churn and protect long-term customer value.

4. Approach: Data Analytics

The analytics approach follows a structured, sequential pipeline that moves from raw data to actionable churn predictions, ensuring that every stage of understanding builds on the one before it:





5. EDA Techniques to Implement

S.No	Technique	Purpose	Visualization / Method
1	Dataset Overview	Understand structure, size, and data types of the churn dataset	df.info(), df.describe(), shape summary
2	Data Quality Assessment	Identify inconsistencies, invalid entries, and formatting issues	Data profiling report
3	Descriptive Statistics	Summarize central tendency and spread of numeric attributes	Mean, median, std-dev tables
4	Missing Value Analysis	Detect and quantify missing customer records or fields	Missingness heatmap, bar chart
5	Duplicate Record Analysis	Identify duplicate customer entries that could bias results	Duplicate count summary
6	Outlier Detection	Detect abnormal values in charges, tenure, and balance	Boxplots, Z-score / IQR method
7	Univariate Analysis	Examine distribution of individual features (e.g., tenure)	Histograms, count plots
8	Bivariate Analysis	Study relationship between one feature and churn	Grouped bar charts, violin plots
9	Multivariate Analysis	Examine interactions among multiple features simultaneously	Pair plots, grouped heatmaps
10	Correlation Analysis	Measure linear relationships between numeric attributes	Correlation heatmap (Pearson)
11	Customer Segmentation Analysis	Group customers by behavior, tenure, or value	Cluster plots, segment profiles
12	Churn Pattern Analysis	Study churn rate across contract type, tenure, and	Stacked bar charts, trend lines

S.No	Technique	Purpose	Visualization / Method
		charges	
13	Feature Engineering	Construct new behavior-based predictive attributes	Derived columns, ratio features
14	Feature Selection	Select the most predictive attributes for modeling	Correlation filter, feature importance

6. Preprocessing

- Missing value handling — imputation or removal based on feature relevance
- Duplicate record removal to prevent bias in model training
- Categorical encoding (One-Hot / Label Encoding) for attributes such as Contract, Geography, and Payment Method
- Feature scaling (StandardScaler / MinMaxScaler) for numeric attributes such as MonthlyCharges and Balance
- Data transformation to correct skewed distributions where necessary
- Class imbalance handling using SMOTE (Synthetic Minority Over-sampling Technique)
- Train-test split (typically 80:20 or 70:30) with stratification on the churn label

7. Classification Models

Five classification algorithms are trained and compared under identical preprocessing conditions:

- Logistic Regression — a fast, interpretable linear baseline for binary churn classification
- Decision Tree — captures non-linear, rule-based splits in customer behavior
- Random Forest — an ensemble of decision trees that reduces overfitting and improves generalization
- XGBoost — a gradient-boosted ensemble known for strong performance on structured/tabular data
- CatBoost — a gradient-boosted model with efficient native handling of categorical churn attributes

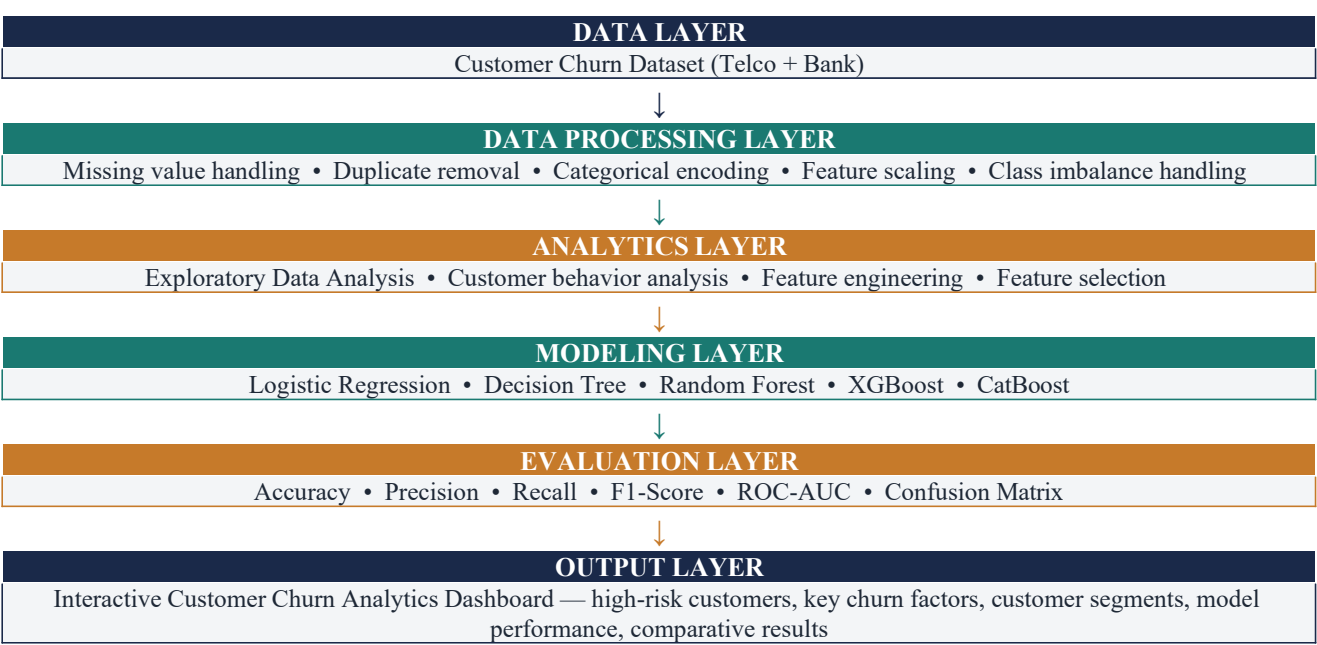
Comparing multiple models rather than relying on a single algorithm is important because churn datasets vary in linearity, feature interactions, and class balance; a multi-model comparison identifies which algorithm generalizes best on the specific data distribution, rather than assuming any one model is universally optimal.

8. Visualization

- Churn distribution charts (overall churn rate, churn by segment)
- Correlation heatmap of numeric attributes
- Boxplots for tenure, monthly charges, and account balance
- Histograms of feature distributions
- Customer segment analysis charts
- Feature importance graph (from tree-based models)
- Confusion matrix for each classifier
- ROC curve and AUC comparison across models

- Precision-Recall curve for imbalance-sensitive evaluation
- Model comparison charts (bar charts across metrics)
- Interactive Streamlit dashboard summarizing all findings

9. Architecture Diagram



10. Comparative Analysis

All five classification models are compared on the held-out test set using the following metrics, with particular emphasis on Recall and F1-Score given the cost of missing an actual churner:

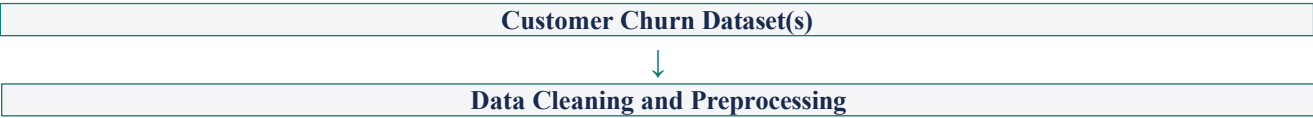
- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Comparison Matrix Template

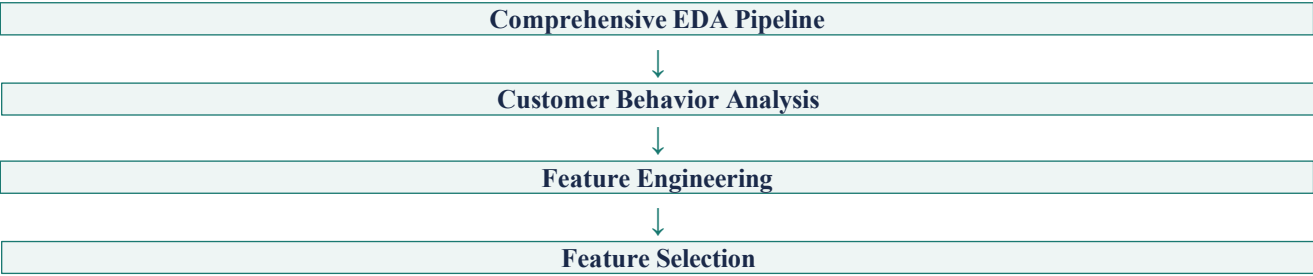
Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	—	—	—	—	—
Decision Tree	—	—	—	—	—
Random Forest	—	—	—	—	—
XGBoost	—	—	—	—	—
CatBoost	—	—	—	—	—

11. Research Pipeline

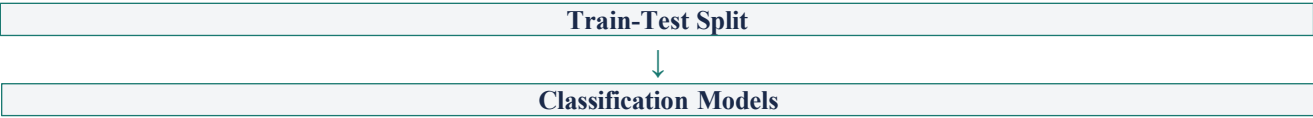
PHASE 1 — Data Collection



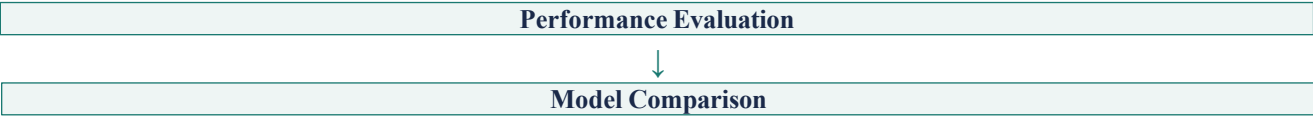
PHASE 2 — Exploratory Data Analysis



PHASE 3 — Modeling



PHASE 4 — Evaluation



PHASE 5 — Visualization and Output



12. Implementation Roadmap

Week	Activities	Deliverables
Week 1	Data collection, cleaning, and preprocessing of both churn datasets	Cleaned and preprocessed dataset ready for analysis
Week 2	Complete Exploratory Data Analysis and customer churn pattern analysis	EDA report with visualizations and key behavioral insights
Week 3	Feature engineering, feature selection, and classification model training	Engineered feature set and trained baseline models
Week 4	Model evaluation, hyperparameter	Comparison matrix and best-
Week	Activities	Deliverables
	tuning, and comparative analysis	performing model identified
Week 5	Dashboard development and final documentation	Interactive churn dashboard and final project report

13. Technology Stack

Category	Tools / Libraries
Data Processing	Pandas, NumPy
EDA and Visualization	Matplotlib, Seaborn, Plotly
Machine Learning	Scikit-learn, XGBoost, CatBoost
Class Imbalance Handling	Imbalanced-learn (SMOTE)
Dashboard	Streamlit
Development Environment	Jupyter Notebook / Google Colab